SMART PANTRY MANAGER

Mphoentle Thomas | Student No. 402411462 | Mobile App Development 700

TABLE OF CONTENTS

Section	Page
1. Introduction	1
2. System design	1
3. Screenshots of the running app	3
4. Key code snippets	12
5. Challenges and solutions	13
6. Conclusion and reflection	13
7. References	14

1. INTRODUCTION

Smart Pantry Manager helps people use ingredients they already have at home. It stores pantry items and compares them with a collection of recipes. This can help users keep track of food and reduce waste.

The main rule is strict recipe matching. The app suggests a recipe only when every required ingredient is in the pantry in a sufficient quantity. A recipe with a missing ingredient is not shown as ready to cook. The app stores its data locally in SQLite and works without an internet connection.

2. SYSTEM DESIGN

2.1 Screen flow

The pantry is the starting screen. The user can add an ingredient, tap a row to edit it, or long-press a row to delete it. The Recipes tab opens the suggestions list. Tapping a suggested recipe opens its details. Settings is available from the bottom navigation.

Screen	Action and destination

Pantry List	Add opens the form; tap a row to edit; long-press to delete; Recipes opens suggestions; Settings opens preferences.
Add/Edit Ingredient	Validate the fields, save the item to SQLite, and return to the pantry.
Suggested Recipes	Choose a fully matched recipe to open its detail screen.
Recipe Detail	Read the ingredients and numbered method.
Settings	Save expiry alert and unit preferences.

2.2 Database design

The SQLite database has three tables. Pantry rows store the user's current ingredients. Recipes store each recipe name and method. Recipe ingredients store the ingredients and amounts needed for each recipe.

PANTRY	RECIPES	RECIPE_INGREDIENTS
id (primary key) name quantity unit expiry_date	id (primary key) name steps	id (primary key) recipe_id (foreign key) name quantity unit
One row per pantry item	One row per recipe	Many ingredient rows for each recipe

One recipe can have many recipe-ingredient rows. The matcher compares these rows with the pantry items.

2.3 Main app components

Component	What it does
DatabaseHelper.java	Creates and upgrades the SQLite tables, loads the recipe

	collection, and reads or changes pantry data.
PantryAdapter.java	Shows pantry items in a RecyclerView and handles taps and long-presses.
RecipeMatcher.java	Compares every recipe ingredient with the pantry, including common plural names and supported unit conversions.
Activities and Intents	Open the pantry, form, suggestions, recipe details, and settings screens. Intents pass item and recipe IDs between screens.

3. SCREENSHOTS OF THE RUNNING APP

These screenshots show the app running on a phone.

Figure 1. Pantry list

The pantry contains butter, eggs, and salt.

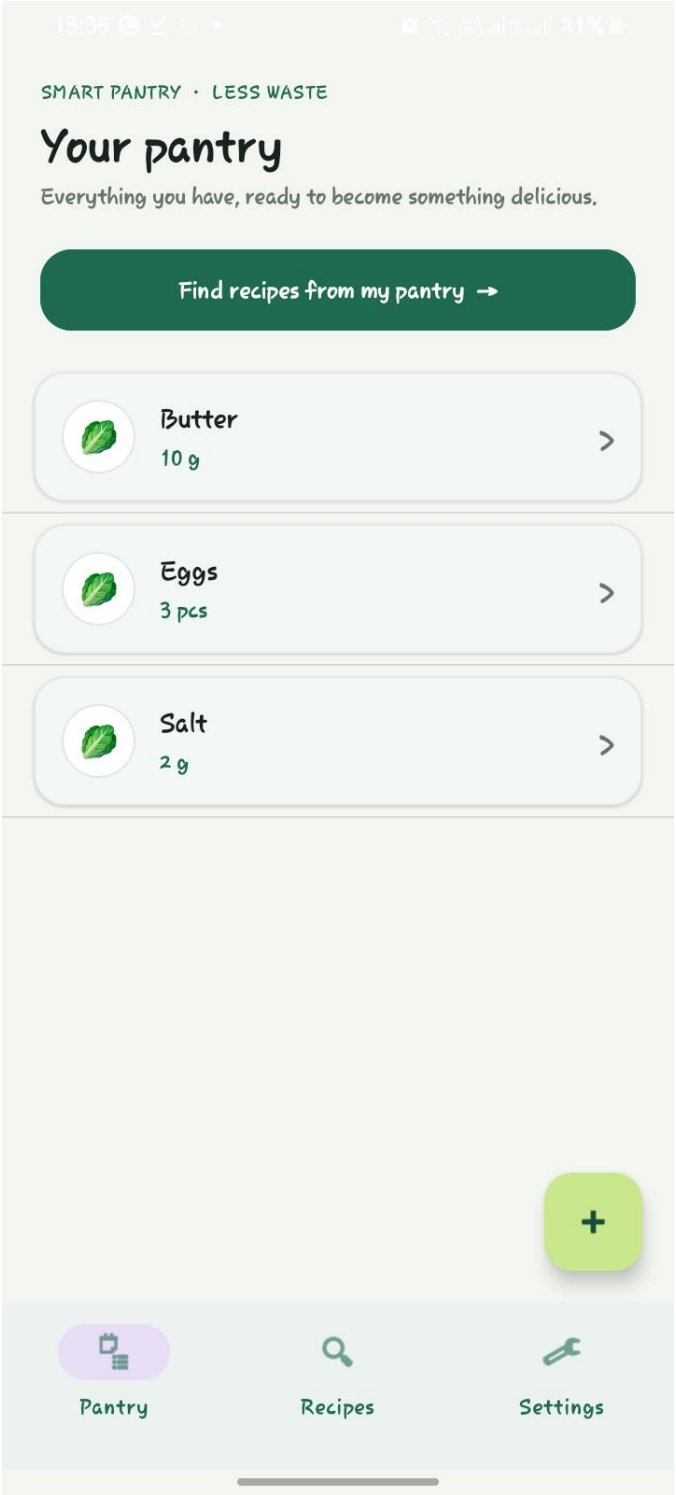

SMART PANTRY · LESS WASTE
Your pantry
Everything you have, ready to become something delicious.
Find recipes from my pantry →
Butter
10 g
Eggs
3 pcs
Salt
2 g
Pantry
Recipes
Settings

Figure 2. Add ingredient form

The form used to add a pantry item.

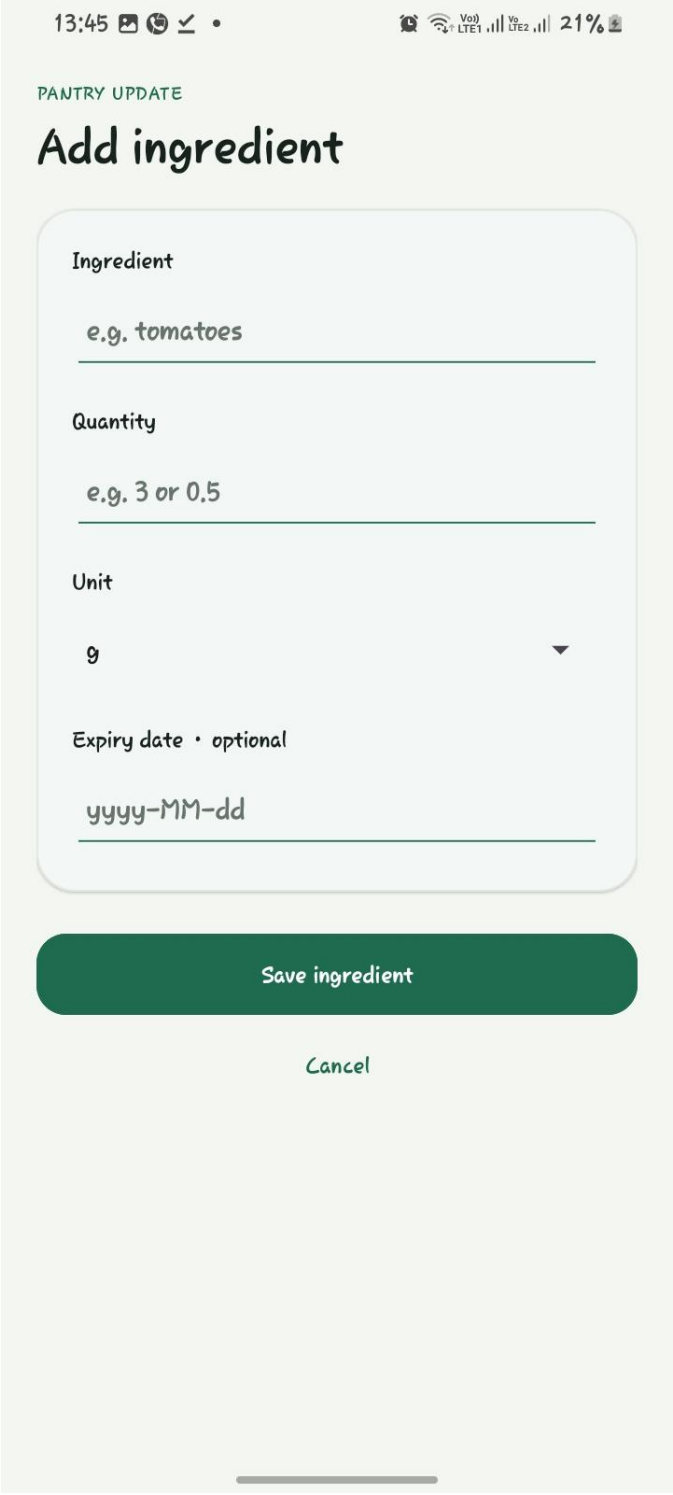


Figure 3. Input validation

An invalid expiry date causes the form to show a validation message.

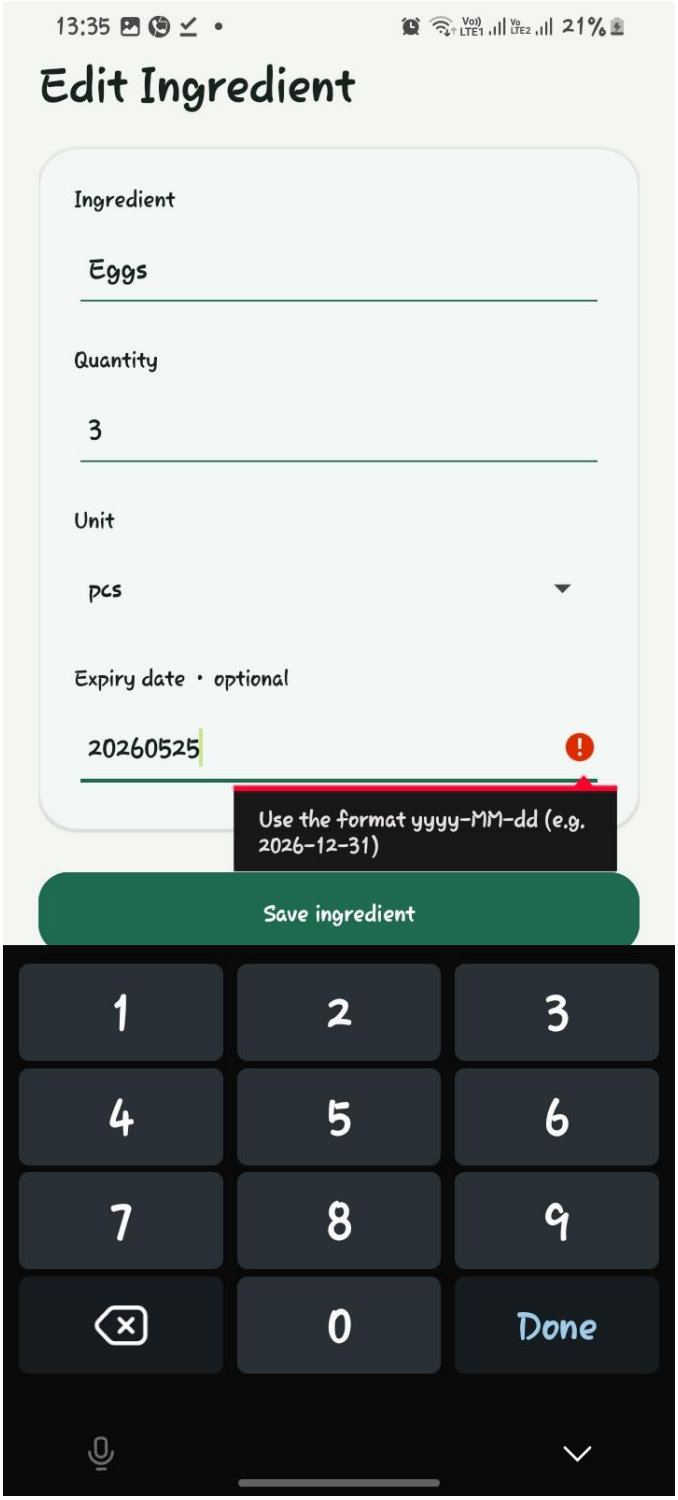


Figure 4. Edit ingredient

An existing pantry item is open with its saved values.

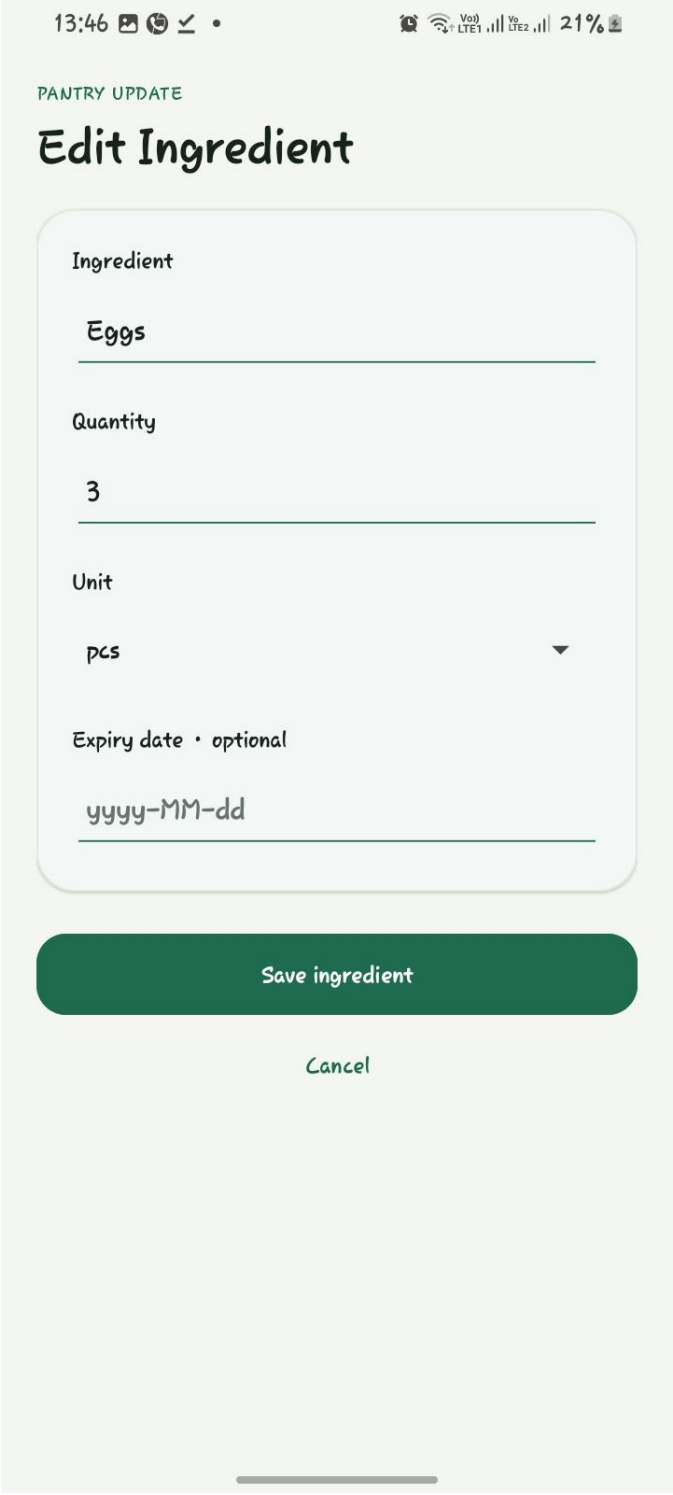


Figure 5. Delete confirmation

The app asks before removing an ingredient.

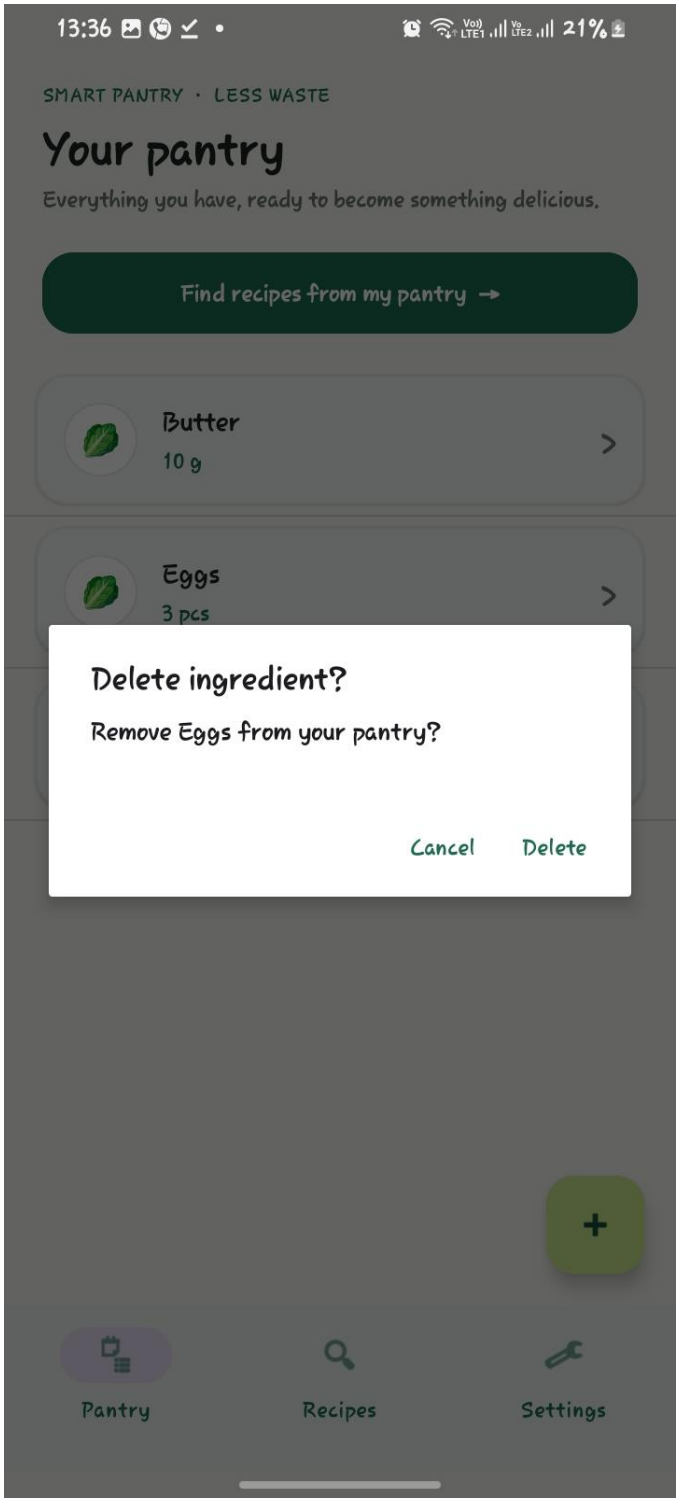


Figure 6. Strict recipe suggestion

Scrambled Eggs appears when the pantry has all required ingredients in sufficient amounts.

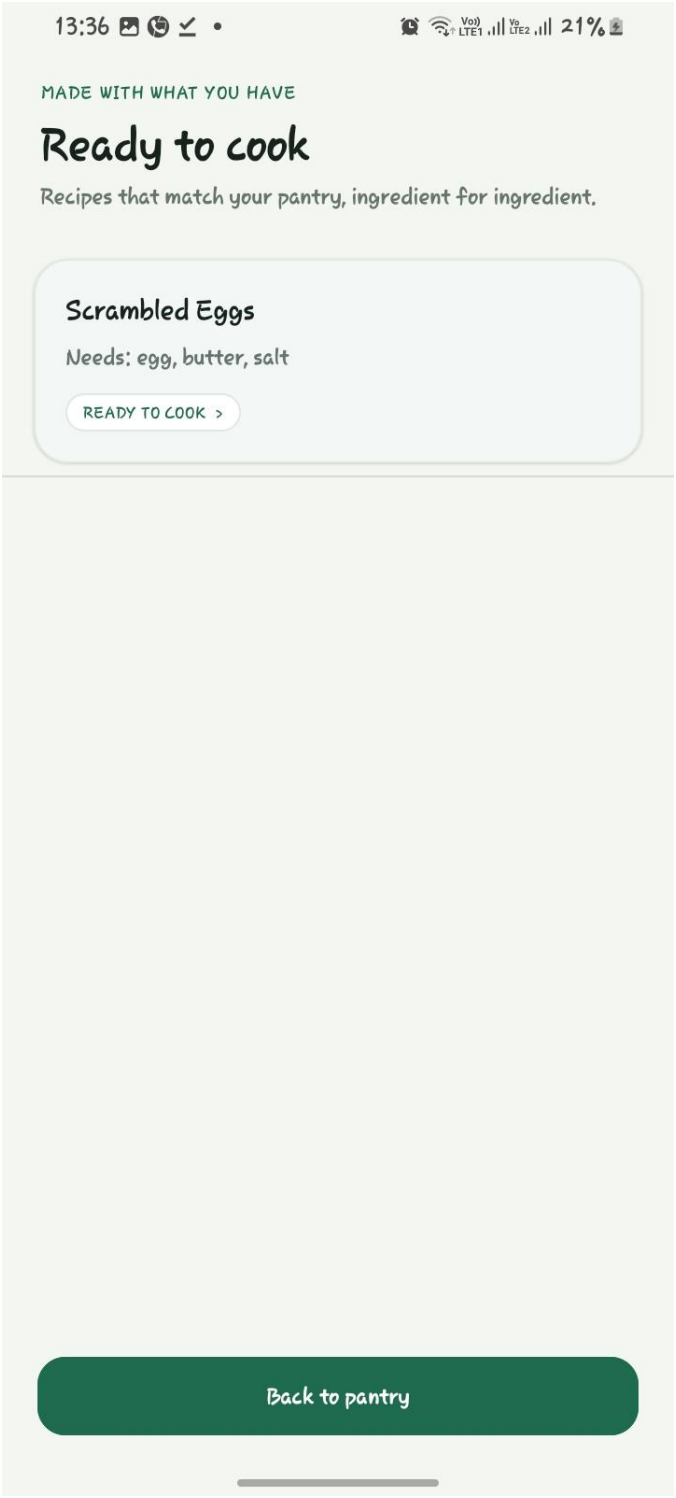


Figure 7. No matching recipes

The app explains that no recipe matches the current pantry.

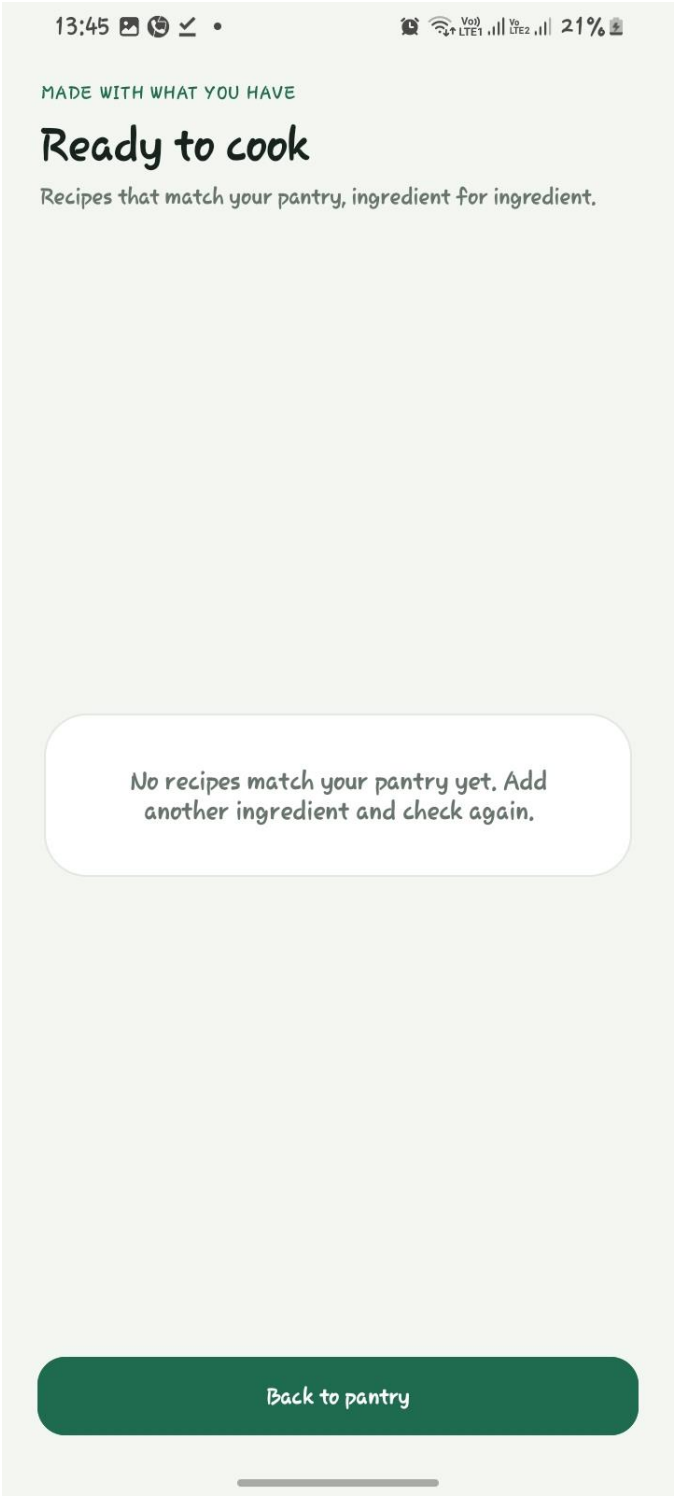
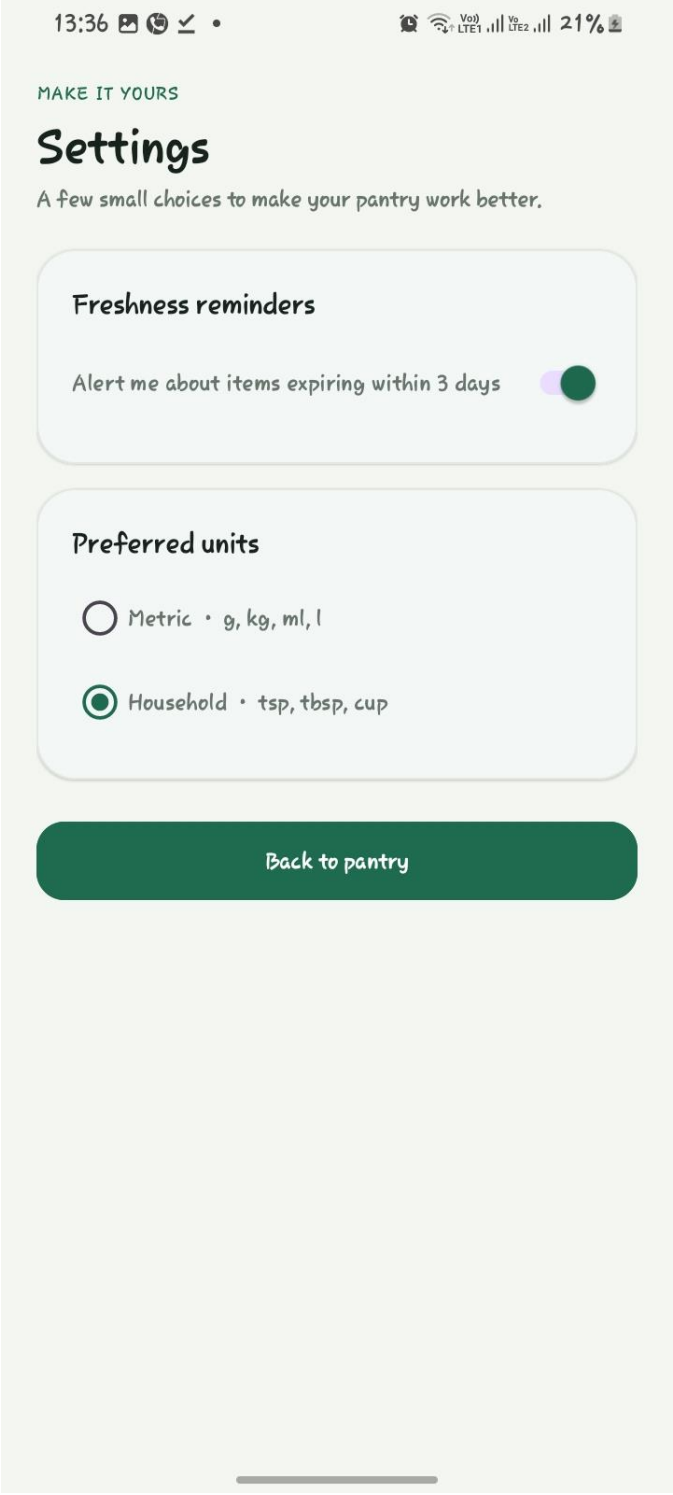


Figure 8. Settings

Expiry alerts and preferred units are shown in Settings.



4. KEY CODE SNIPPETS

These short examples show how the app matches recipes, saves data, opens screens, and refreshes the pantry.

4.1 Strict recipe matching — RecipeMatcher.java

```
for (RecipeIngredient ing : recipe.getIngredients()) {
    String key = makeKey(ing.getName(), ing.getUnit());
    double needed = toBaseAmount(ing.getQuantity(), ing.getUnit());
    Double have = available.get(key);
    if (have == null || have + 0.0001 < needed) return false;
}
return true;
```

The method checks every required ingredient. It rejects the recipe if an item is missing or the pantry has too little. It suggests the recipe only if every check passes.

4.2 Saving pantry data — DatabaseHelper.java

```
ContentValues values = new ContentValues();
values.put(COL_NAME, item.getName());
values.put(COL_QTY, item.getQuantity());
values.put(COL_UNIT, item.getUnit());
values.put(COL_EXPIRY, item.getExpiryDate());
db.insert(TABLE_PANTRY, null, values);
```

The helper stores the ingredient fields in the pantry table. The app also has methods to read, update, and delete pantry records.

4.3 Opening the edit screen — MainActivity.java

```
Intent intent = new Intent(MainActivity.this, AddEditActivity.class);
intent.putExtra(AddEditActivity.EXTRA_ITEM_ID, item.getId());
startActivity(intent);
```

The Intent opens the form and passes the selected pantry item's ID. The form uses that ID to load and update the correct row.

4.4 Refreshing the pantry — MainActivity.java

```
protected void onResume() {
    super.onResume();
    loadPantry();
    bottomNav.setSelectedItemId(R.id.nav_pantry);
    showExpiryAlert();
}
```

Android calls onResume when the pantry screen becomes active again. The app reloads the list so saved changes appear.

5. CHALLENGES AND SOLUTIONS

The first challenge was making strict matching handle simple differences such as singular and plural names and different units. RecipeMatcher converts supported units to base units and normalizes common names before comparing quantities.

The second challenge was an empty recipe list on an app that had already been installed before recipe seeding was fixed. The database version was increased. On upgrade, the app now seeds recipes only when the recipe table is empty and keeps existing pantry items.

6. CONCLUSION AND REFLECTION

Smart Pantry Manager combines Android screens, Intents, RecyclerView, input validation, SharedPreferences, and SQLite. Its strict matcher ensures that every suggested recipe can be made with the ingredients and quantities already in the pantry.

The project demonstrates how a local database and clear matching rules can support a useful everyday task. A future version could add more recipes, richer ingredient matching, and clearer expiry-date warnings.

7. REFERENCES

Google (n.d.) SQLiteOpenHelper. Android Developers API reference. Available at: https://developer.android.com/reference/android/database/sqlite/SQLiteOpenHelper (Accessed: 4 November 2026).

Google (n.d.) Create dynamic lists with RecyclerView. Android Developers. Available at: https://developer.android.com/develop/ui/views/layout/recyclerview (Accessed: 4 November 2026).

Google (n.d.) Intents and intent filters. Android Developers. Available at: https://developer.android.com/guide/components/intents-filters (Accessed: 4 November 2026).

Richfield Graduate Institute of Technology (2026) Mobile App Development 700: Practical Assignment — Smart Pantry Manager. Assignment brief supplied for the module.

Smart Pantry Manager source code (2026). GitHub repository: https://github.com/Saucydotnineseven/SmartPantry